

Wide-Angle Multislice for Electron Microscopy via the Wave Propagation Method

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Abstract

Multislice simulation is the standard method for modeling electron beam-sample interactions in transmission electron microscopy (TEM). The traditional multislice algorithm is calculated using a split-step Fourier method that employs a paraxial (Fresnel) propagator, limiting its accuracy to small scattering angles (typically $< 7^\circ$) and ignoring the change of the electron wavelength inside the sample potential. These limitations can be significant in scenarios involving high-angle scattering or thick, high-Z samples. In this work, we present a systematic comparison of four propagation methods of increasing physical rigour: Fresnel multislice, angular spectrum multislice, the Wave Propagation Method (WPM) from the light optics community, and a Krylov-subspace (Lanczos) forward solver for the Klein-Gordon equation. We cast the Dirac, Klein-Gordon, and relativistically corrected Schrödinger equations into a common Helmholtz form, adapt the WPM to electron microscopy via a differentiable refractive-index binning strategy, and benchmark all four methods against high-angle reference solutions across z-sampling convergence, convergent-beam diffraction at probe semi-angles up to 100 mrad, and frozen-phonon Si [111] simulations. Our principal finding is that replacing the Fresnel propagation kernel with the exact angular spectrum kernel—a minimal, inexpensive modification to standard multislice—is sufficient to capture high-angle scattering with accuracy comparable to the far more computationally expensive WPM and Klein-Gordon solvers across all tested regimes.

1 Introduction

The traditional multislice propagation method is the standard computational tool [1, 2] to solve the time-independent relativistically corrected Schrödinger wave equation. By assuming a mean-field independent-atom potential and splitting the three-dimensional object into thin slices while alternating between transmission and propagation steps, the multislice method allows for efficient inclusion of dynamical scattering effects essential for interpreting modern electron microscopy data. However, the standard implementation of the multislice algorithm relies on the Fresnel propagator whose formulation inherently assumes the paraxial approximation, where scattering angles are small (typically $< 7^\circ$).

While this approximation holds for many conventional imaging conditions, it becomes increasingly inaccurate in regimes that contain high-angle scattering effects. Specifically, discrepancies arise in scenarios involving transmission through thick, high-atomic-number (high-Z) samples, where the cumulative scattering angles exceed the limits of the paraxial assumption. Previous work by Rother and Scheerschmidt [3] have demonstrated in the case of a periodic thick gold sample that spin effects included in the Dirac wave equation and the complete relativistic factor of the Klein-Gordon equation are mostly negligible when comparing the amplitude of high-angle beams in a thick gold sample. However, the paraxial nature of multislice begins to show deviations from a complete solution to the wave equation after five unit cells of thickness. Their conclusion was that the linearization of the wave equation (the paraxial approximation) leads to significant divergence in diffraction amplitudes at high angles (e.g., 135 mrad) when propagating through thick crystals.

Methods to overcome the paraxial approximation in solutions to the wave equation have already been developed in the light optics community (where the multislice method is known as the Beam Propagation Method (BPM)) by developing non-paraxial propagation techniques through the use of higher order Taylor expansions of the propagation operator of the time-independent Helmholtz wave equation, known as Padé approximants [4]. Padé approximants have been shown to be accurate to

much higher angles than the traditional Fresnel propagator multislice method, but can become computationally prohibitive to compute for higher order approximants. In order to overcome the limitations of the paraxial approximation and the computational cost of higher order Padé approximants, the Wave Propagation Method (WPM) was developed by [5], which solves the Helmholtz wave equation in the one-way wave approximation using a Fourier integral formulation.

The WPM has been demonstrated to be accurate up to 85 degrees from the optical axis [6] and can leverage the efficiency of the Fast Fourier Transform (FFT) for computation, meaning it can be implemented in a way that is more computationally efficient than higher order Padé approximants [4]. Given that the WPM was developed as a solution for light optics and the time-independent Helmholtz wave equation, we first establish the connection between the time-independent Dirac equation, Klein-Gordon equation, and the approximated Klein-Gordon equation (a.k.a. relativistically corrected Schrodinger equation) in the framework of the Helmholtz wave equation in order to justify the application of the WPM to electron microscopy multislice simulations.

In this paper, we make three contributions:

- We cast the Dirac, Klein-Gordon, and relativistically corrected Schrödinger equations into a common Helmholtz form to identify the effective electron refractive index.
- We adapt the WPM to electron microscopy, derive the linearization that leads to the Fresnel multislice propagator, and introduce a differentiable refractive-index binning strategy that makes the WPM computationally tractable.
- We present a systematic numerical comparison—across z-sampling convergence, probe convergence angle, and frozen-phonon simulations—showing that the angular spectrum propagation kernel alone captures the high-angle accuracy of the more expensive WPM and Klein-Gordon solvers.

The remainder of the paper reviews the relevant wave equations, derives WPM and its relationship to multislice, presents the binned implementation, and evaluates accuracy across four progressively complex numerical experiments.

2 Relativistic Dirac Equation and the approximated Klein-Gordon Equation

As described by [3], the relativistic Dirac equation is the starting point for the most accurate description of high-energy electron scattering:

$$E\Psi(\vec{r}) = [c\vec{\alpha} \cdot \hat{p} + m_0c^2\beta + V(\vec{r})]\Psi(\vec{r}), \quad (1)$$

where E denotes the total energy, c the velocity of light, m_0 the electron rest mass, V the electrostatic potential energy (with $V < 0$ for an attractive nuclear potential), $\vec{\alpha}$ and β the Dirac matrices, $\hat{p}$ the momentum operator and $\vec{r}$ the three-dimensional space coordinate. A solution to equation (1) was used by [3] as a "ground-truth" numerical baseline to establish the accuracy of the multislice solution to the relativistically corrected Schrodinger equation. In their work, they demonstrated that:

1. The spin effects of the Dirac equation are mostly negligible to the error in amplitude of a central and high angle beam scattered through a thick sample of gold.
2. The Taylor expansion that is applied to the $E^2 - V^2$ term, to enable the use of the simple relationship σV_{elec} , where V_{elec} is the electrostatic potential in Volts and σ is the so-called interaction constant given by:

$$\sigma = \frac{2\pi\gamma m_0 e \lambda}{h^2}$$

with γ the Lorentz factor, e the elementary charge, and λ the relativistic de Broglie wavelength, does not contribute a significant error to the solution of the wave equation.

However, for a diffracted beam at 135 mrad and scattering through a thick sample of gold, they found a divergence between the multislice solution solved via a split-step Fourier method with the

Fresnel propagator and a "ground-truth" ODE solution to the approximated Klein-Gordon (a.k.a. relativistically corrected Schrodinger wave equation), Klein-Gordon, and Dirac equation. They explain the discrepancy by the fact that the multislice solution linearizes the second order term in the wave equation, $d^2\psi/dz^2$, by assuming that the wave function can be written as a slowly varying envelope modulating a fast oscillating term along the beam propagation direction z . Furthermore, we note that the multislice solution ignores the change of wavelength of the electron as it propagates through a potential $V(\mathbf{r})$, since MS splits up the potential interaction and free-space propagation steps. In general, we cannot overcome these limitations of the multislice method by just solving the wave equation in an ODE form, as it normally requires a periodic potential, and the slow speed of an ODE solution to the wave equation does not lend itself to repeated simulations over many frozen phonon configurations to include thermal vibrations, which has become a standard requirement to include thermal diffuse scattering in TEM simulations. A method that is both accurate to high angles and can leverage the efficiency of the fast Fourier transform is thus desirable to overcome these limitations.

In order to investigate non-paraxial solutions to the relativistically corrected Schrodinger equation that can leverage the efficiency of the fast Fourier transform, it is insightful to write the Dirac, Klein-Gordon, and approximated Klein-Gordon equations in the same framework as a time-independent Helmholtz-like wave equation, which is of the form:

$$\nabla^2 u + k^2 n^2(\mathbf{r}) u = 0. \quad (2)$$

with u the wave function, k the wave number in vacuum, and $n(\mathbf{r})$ the refractive index distribution of the medium.

Starting from the time-independent Dirac equation:

$$\left[c\vec{\alpha} \cdot \hat{\mathbf{p}} + \beta m_0 c^2 + V \right] \psi = E\psi \quad (3)$$

by applying the operator $(c\vec{\alpha} \cdot \hat{\mathbf{p}})$, and performing some algebraic manipulations, we arrive at:

$$\nabla^2 \psi + k_0^2 n^2(\mathbf{r}) \psi = -\frac{i}{\hbar c} (\vec{\alpha} \cdot \nabla V) \psi \quad (4)$$

where

$k_0 = \frac{\sqrt{E^2 - m_0^2 c^4}}{\hbar c}$ represents the vacuum wave number, and $n(\mathbf{r}) = \sqrt{\frac{(E-V)^2 - m_0^2 c^4}{E^2 - m_0^2 c^4}}$ is the effective refractive index of the medium.

From this Helmholtz-like representation, it is straightforward to arrive at the Klein-Gordon equation by setting the α matrices to zero and ignoring the effects of spin,

$$\nabla^2 \psi + k_0^2 n^2(\mathbf{r}) \psi = \nabla^2 \psi + \frac{(E-V)^2 - m_0^2 c^4}{\hbar^2 c^2} \psi = 0. \quad (5)$$

The relativistically corrected Schrodinger equation arises from the Klein-Gordon equation by assuming that the potential V is small compared to the total energy E of the electron, and that when $(E-V)^2$ is expanded, the V^2 term can be neglected giving:

$$\nabla^2 \Psi + \left[\frac{E^2 - m_0^2 c^4}{\hbar^2 c^2} - \frac{2\gamma m_0 V}{\hbar^2} \right] \Psi = 0 \quad (6)$$

Writing this in Helmholtz form $\nabla^2 \Psi + k_0^2 n^2(\mathbf{r}) \Psi = 0$ identifies the effective refractive index for the relativistically corrected Schrödinger equation:

$$n(\mathbf{r}) = \sqrt{1 - \frac{2\gamma m_0 V}{\hbar^2 k_0^2}} \quad (7)$$

which forms the third row of Table 1.

Writing equation (1) in its Helmholtz form (4) allows us to clearly see the relationship between the time-independent Dirac equation and its approximations and the classical Helmholtz wave equation in optics. Understanding the connections in this way allows us to directly apply non-paraxial propagation methods from the optics community that solve the Helmholtz equation to electron microscopy, provided we use the correct refractive index function for electrons in a potential.

Equation	Helmholtz Form	Refractive Index ($n(\mathbf{r})$)
Dirac	$\nabla^2\psi + k_0^2 n^2\psi = -\frac{i}{\hbar c}(\vec{\alpha} \cdot \nabla V)\psi$	$\sqrt{\frac{(E-V)^2 - m_0^2 c^4}{E^2 - m_0^2 c^4}}$
Klein-Gordon	$\nabla^2\psi + k_0^2 n^2\psi = 0$	$\sqrt{\frac{(E-V)^2 - m_0^2 c^4}{E^2 - m_0^2 c^4}}$
Rel. Schrödinger	$\nabla^2\psi + k_0^2 n^2\psi = 0$	$\sqrt{1 - \frac{2\gamma m_0 V}{\hbar^2 k_0^2}}$

Table 1: Helmholtz versions of quantum mechanics equations. The vacuum wave number is defined as $k_0 = \frac{\sqrt{E^2 - m_0^2 c^4}}{\hbar c}$, γ is the Lorentz factor, and m_0 is the electron rest mass.

3 Wave Propagation Method

The Wave Propagation Method (WPM) is a numerical method for solving the Helmholtz wave equation in the forward scattering approximation that was introduced by [5] and has been demonstrated to be accurate up to 85 degrees from the optical axis [6]. A systematic derivation of the WPM from the Helmholtz wave equation is given by [6]. For conciseness, we present a summary of the derivation here.

Starting with the scalar Helmholtz equation for the wavefunction ψ in a medium with a z-dependent refractive index n :

$$(\partial_z^2 + \nabla_\perp^2 + k_0^2 n^2(x, y, z))\psi = 0 \quad (8)$$

By neglecting back-scattering (the "one-way" wave approximation), the evolution of the wavefunction is governed by the square-root operator:

$$\partial_z \psi = i\sqrt{\nabla_\perp^2 + k_0^2 n^2} \psi \quad (9)$$

which has a solution according to:

$$\psi(x, y, z + \Delta z) = e^{i\sqrt{\nabla_\perp^2 + k_0^2 n^2}\Delta z}\psi(x, y, z) \quad (10)$$

From this point, one can derive the typical multislice solution (BPM in optics literature) by applying various approximations to the square-root operator. Standard multislice approximates this square root by splitting the Laplacian ($\nabla_\perp^2$) and refractive index (n^2) terms. WPM, however, solves the propagation step Δz using a more rigorous Fourier integral formulation:

$$\psi(x, y, z + \Delta z) = \frac{1}{(2\pi)^2} \iint \tilde{\psi}(k_x, k_y, z) e^{i\sqrt{k_0^2 n^2(x, y) - k_\perp^2}\Delta z} e^{i(k_x x + k_y y)} dk_x dk_y \quad (11)$$

Here, $\tilde{\psi}$ is the Fourier transform of the wavefunction.

Crucially, the longitudinal wavevector $k_z = \sqrt{k_0^2 n^2 - k_\perp^2}$ is dependent on both real space coordinates (x, y) and reciprocal space coordinates (k_x, k_y) , and includes any modification to the wavelength inside the medium via the refractive index $n(x, y)$.

Equation (11) can take advantage of the Fast Fourier Transform (FFT) for efficient computation:

$$\psi(x, y, z + \Delta z) = \mathcal{F}^{-1} \left[e^{ik_z(x, y)\Delta z} \cdot \mathcal{F}\{\psi(x, y, z)\} \right] \quad (12)$$

and is implemented in a slice-by-slice manner akin to multislice. When the refractive index is constant across the transverse plane, WPM reduces to the Angular Spectrum of Plane Waves method of propagation.

3.1 Linearization and the Fresnel multislice propagator

To see why the Angular Spectrum Method (ASM) does not fix multislice, start from the one-way operator in equation (9). In Fourier space, the exact longitudinal wavevector is

$$k_z = \sqrt{k_0^2 n^2 - k_\perp^2}, \quad k_\perp^2 = k_x^2 + k_y^2, \quad (13)$$

and the exact propagator $\exp(ik_z \Delta z)$ is precisely the ASM kernel for a homogeneous slice. Multi-slice instead linearizes the square-root operator by assuming $k_\perp^2 \ll k_0^2 n^2$ and a slowly varying envelope. Expanding the operator in real space gives

$$\sqrt{k_0^2 n^2 + \nabla_\perp^2} = k_0 n \sqrt{1 + \frac{\nabla_\perp^2}{k_0^2 n^2}} \approx k_0 n + \frac{\nabla_\perp^2}{2k_0 n} - \frac{\nabla_\perp^4}{8k_0^3 n^3} + \dots \quad (14)$$

Retaining only the first two terms yields the paraxial equation

$$\partial_z \psi \approx ik_0 n \psi + \frac{i}{2k_0 n} \nabla_\perp^2 \psi. \quad (15)$$

For weak scattering, $n = 1 + \delta n$ with $|\delta n| \ll 1$, and replacing n in the diffraction term by unity gives

$$\psi(x, y, z + \Delta z) \approx e^{ik_0(n-1)\Delta z} \mathcal{F}^{-1} \left[e^{-i \frac{k_\perp^2}{2k_0} \Delta z} \mathcal{F}\{\psi(x, y, z)\} \right], \quad (16)$$

which is the split-step Fresnel multislice propagator (up to a global phase). This linearization both truncates higher-order terms and assumes a commutation between the potential phase and diffraction operator. As a result, swapping in an ASM kernel after the split-step does not undo the paraxial approximation; wide-angle accuracy requires retaining higher-order terms (Padé approximants) or using WPM, which keeps the square-root operator intact.

Although the WPM is accurate to high angles, its main drawback is that for each unique refractive index value in the slice, a unique propagator must be computed. Without any modification, the order of computational complexity is $O(N^{2D})$ where N is the number of grid points, and D is the number of transverse dimensions. Such a computational cost is prohibitive, as an object slice that is discretized into, for example, 100×100 pixels and has a unique refractive index value at each pixel would require 10,000 unique Fourier transforms per slice for one propagation step.

[6] proposed a modification to the WPM that reduces the computational complexity from $O(N^{2D})$ to $O(MN^D \log N)$, where M is the number of "unique" refractive index values in the slice. By applying a mask $I_m(x, y)$ that selects pixels with the same refractive index n_m , the propagation step can be written as a sum over M unique propagations. However, in the case of electron microscopy multislice this algorithm poses two problems: First, applying a mask to the propagator negates the possibility to differentiate through the propagation step, which is a necessary requirement to use the WPM as a forward model to solve inverse problems via gradient descent, as is needed to perform ptychography. Second, in the case of electron microscopy, the refractive index is a continuous function of the potential $V(\mathbf{r})$, and thus there are usually not very many "unique" refractive index values in a slice. [7] attempted to overcome both of these problems by introducing a continuous binning step in a binary refractive index medium.

4 Binning of refractive index values for efficient WPM multislice

To make the computation of the wave propagation method tractable and limit the need for $O(N^{2D})$ computations, we adapt the WPM algorithm presented in [7] which uses a refractive index binning strategy to limit the number of unique refractive index values in a slice while maintaining differentiability with respect to the refractive index map $n(x, y)$.

To implement this method, we first discretize the refractive index range $[n_{\min}, n_{\max}]$ of the current slice into N_{bins} discrete reference values, denoted as $\{n_{\text{ref}}^{(k)}\}_{k=1}^{N_{\text{bins}}}$.

In a typical multislice potential, many of the refractive index values will be very close to 1 (vacuum region), while the refractive index values of an atom will concentrate in a narrow range slightly above 1. Since there are likely to be many more refractive index values near the vacuum region, we employ a quadratic distribution for the reference indices rather than using linearly spaced bins in order to

allocate more bins near $n = 1$. The reference indices are distributed according to a quadratic function determined by the parameter γ , with $\gamma = 2.0$ for quadratic spacing:

$$n_{\text{ref}}^{(k)} = n_{\text{min}} + (n_{\text{max}} - n_{\text{min}}) \left(\frac{k}{N_{\text{bins}} - 1} \right)^\gamma$$

which allows for denser binning near the vacuum refractive index when $\gamma > 1$.

Next, for a propagation step Δz , we compute a bank of propagated fields $u_k(x, y)$. Each field u_k corresponds to the diffraction of the input field $u_{\text{in}}(x, y)$ assuming a homogeneous medium with refractive index $n_{\text{ref}}^{(k)}$. This is performed efficiently via batch FFT operations using the angular spectrum kernel:

$$u_k(x, y) = \mathcal{F}^{-1} \left(\mathcal{F}(u_{\text{in}}) e^{i\Delta z \sqrt{(k_0 n_{\text{ref}}^{(k)})^2 - k_x^2 - k_y^2}} \right)$$

To evaluate the field at a specific spatial coordinate (x, y) where the local refractive index is $n(x, y)$, we identify the two bounding reference bins such that $n_L \leq n(x, y) < n_R$, where $n_L, n_R \in \{n_{\text{ref}}^{(k)}\}$. We adopt a differentiable interpolation scheme by defining a local interpolation weight

$$w_{\text{raw}} = \frac{n(x, y) - n_L}{n_R - n_L}, \quad w_{\text{raw}} \in [0, 1]$$

To ensure smooth gradients and minimize discretization artifacts, we map this linear weight through a smoothstep activation function $S(w)$, akin to the spatial partitioning technique demonstrated by [8] via a cubic Hermite interpolation function:

$$w = S(w_{\text{raw}}) = 3w_{\text{raw}}^2 - 2w_{\text{raw}}^3$$

While linear interpolation would satisfy the requirement for differentiability, it could result in discontinuities in the gradient at the bin boundaries in a gradient-based optimization scheme.

The final field at the next step $u(x, y, z + \Delta z)$ is computed as the weighted superposition of the fields propagated at the bounding indices:

$$u(x, y) = (1 - w) \cdot u_L(x, y) + w \cdot u_R(x, y)$$

This formulation ensures that the propagation model is fully differentiable with respect to the refractive index map $n(x, y)$, enabling inverse reconstruction of potentials via automatic differentiation, and that only a limited number of FFTs (equal to N_{bins}) are required per propagation step, significantly reducing the computational cost compared to a naive WPM implementation. The accuracy of this binned WPM multislice method depends on the number of bins N_{bins} chosen and will change on a case-by-case basis depending on the refractive index distribution of the sample. One can mitigate this potential source of error by performing a test simulation and doubling the number of bins until no difference is observed in the output wavefunction.

In general, the WPM will need a GPU for efficient computation, and we have found that N_{bins} between 64 and 512 are sufficient for the simulations presented in this work. This is a marked improvement over the naive case, as it enables the WPM to simulate many slices with larger grid sizes (512×512 or 1024×1024 pixels) in a reasonable time frame (seconds to minutes), provided the refractive index binning strategy is capable of correctly capturing the refractive index distribution of the sample.

5 Method Summary and Comparison

Figure 1 summarizes the relationship between the exact Helmholtz formulation, the WPM, and Fresnel multislice, highlighting the trade-off between physical accuracy and computational efficiency.

6 Results

We present four numerical experiments of increasing physical complexity, all comparing Fresnel multislice (FMS), angular spectrum multislice (ASMS), the Wave Propagation Method (WPM), and a Krylov–Lanczos forward solver for the Klein-Gordon equation (KG FWD). In each case the ground truth is either a converged ODE solution of the full second-order Klein-Gordon equation (Sections 6.1–6.3) or the KG FWD Lanczos propagator at matched z-sampling (Section 6.4).

6.1 Verification against Rother & Scheerschmidt (2009)

To verify all four propagation methods, we recreated the simulation presented in figure 3 of [3], which tracks the amplitude of a central beam $[0, 0]$ and a high-angle beam $[0, 28]$ at 135 mrad after propagation through a thick gold $[100]$ crystal at 300 keV. The ground truth is a converged ODE solution of the full second-order Klein-Gordon equation.

For the central beam, all methods agree closely. For the high-angle beam at 135 mrad, the Fresnel multislice solution diverges from the ground truth after approximately five unit cells, consistent with the findings of [3] who attributed the discrepancy to the linearization of the square-root operator. However, the angular spectrum multislice solution—which differs from Fresnel MS only by replacing the parabolic propagation kernel $\exp(-ik_{\perp}^2 \Delta z / 2k_0)$ with the exact kernel $\exp(i\sqrt{k_0^2 - k_{\perp}^2} \Delta z)$ —closely tracks the more expensive WPM and KG solutions. This is the first indication that the angular spectrum kernel alone captures the essential high-angle physics. Table 2 quantifies this observation with RMSE values computed over the first 25 unit cells against the digitized paper reference.

Method	RMSE $[0, 0]$	RMSE $[0, 28]$
Fresnel MS	—	—
Angular Spectrum MS	—	—
WPM	—	—
KG FWD Lanczos	—	—
Full KG ODE	—	—

Table 2: RMSE of beam amplitudes versus the digitized reference data of [3] over the first 25 unit cells. Values to be filled from simulation output.

6.2 Z-sampling convergence

Having established qualitative agreement, we now test how each method converges as the z-sampling (number of slices per unit cell) is refined. We propagate a plane wave through ~ 100 unit cells of Au $[110]$ at 200 keV. The ground truth is a KG ODE solution with 128 slices per unit cell and tight tolerances ($\text{rtol} = 10^{-9}$, $\text{atol} = 10^{-11}$). For each method, the z-sampling is varied from 1 to 128 slices per unit cell and the relative L^2 norm of the diffraction pattern difference is computed against the ground truth.

Figure 3 shows that all four methods converge at a similar rate as the z-sampling is refined, and that the WPM and KG FWD Lanczos methods do not permit coarser z-sampling than standard multislice for the same level of accuracy. Beyond ~ 8 slices per unit cell, further refinement yields diminishing returns—less than one order of magnitude improvement in diffraction pattern error despite a large increase in computational cost. The angular spectrum method achieves comparable accuracy to WPM and KG FWD at each z-sampling level, at a fraction of the computational cost.

6.3 Accuracy versus probe convergence angle

Next, we test each method’s accuracy as a function of probe convergence semi-angle for a convergent-beam electron diffraction (CBED) geometry. A convergent probe with semi-angles ranging from 5 to 100 mrad is propagated through the same Au $[110]$ crystal at 200 keV, using 32 slices per unit cell for the test methods and a KG ODE ground truth with 128 slices per unit cell.

Figure 4 plots the relative L^2 diffraction pattern error as a function of semi-angle. Across the entire angular range, the angular spectrum method provides accuracy comparable to both WPM and

KG FWD Lanczos, while the Fresnel propagator shows a growing error at higher convergence angles. Figure 5 displays the diffraction patterns at four representative semi-angles (5, 20, 50, and 100 mrad), confirming that no visually discernible differences exist between angular spectrum MS and the more expensive methods even at 100 mrad.

6.4 Frozen phonon CBED — Si [111]

Finally, we test each method in a realistic frozen-phonon CBED simulation. Si [111] is propagated at 200 keV with a 6.2 mrad convergent probe, 120 unit cells along the beam direction, 16 slices per unit cell, and 16 frozen phonon configurations ($\sigma_{\text{Si}} = 0.078 \text{ \AA}$). For this experiment, the KG FWD Lanczos propagator at matched z-sampling serves as the ground truth, since the full KG ODE with 128 slices per cell is prohibitively expensive for repeated frozen-phonon runs.

Figure 6 shows the frozen-phonon averaged CBED patterns for each method on a logarithmic intensity scale. Figure 7 presents difference maps of each method’s averaged diffraction pattern and exit-wave amplitude relative to the KG FWD reference. Table 3 lists the RMSE, relative L^2 error, and mean wall-clock time per frozen phonon configuration.

Even in this realistic scenario with thermal disorder, the angular spectrum method produces diffraction patterns that are effectively identical to the KG FWD reference, while requiring a small fraction of the compute time. The WPM results are likewise comparable, but at substantially higher cost. These results confirm that across all tested regimes—from single-beam verification to frozen-phonon averaged CBED—the angular spectrum propagation kernel is sufficient to capture the physics of high-angle electron scattering without recourse to more expensive methods.

Method	RMSE	Rel. L^2	Time/config (s)
Fresnel MS	—	—	—
Angular Spectrum MS	—	—	—
WPM	—	—	—
KG FWD Lanczos (ref)	—	—	—

Table 3: Error metrics and runtime for frozen-phonon averaged CBED of Si [111] relative to KG FWD Lanczos. Values to be filled from simulation output.

7 Conclusion

We presented a systematic comparison of four propagation methods for electron multislice simulation—Fresnel multislice, angular spectrum multislice, the Wave Propagation Method, and a Krylov–Lanczos Klein-Gordon forward solver—benchmarked across four experiments of increasing physical complexity: high-angle beam amplitude verification against published results, z-sampling convergence, convergent-beam diffraction at probe semi-angles up to 100 mrad, and frozen-phonon CBED of Si [111].

Our principal finding is that replacing the Fresnel propagation kernel with the exact angular spectrum kernel—a minimal and inexpensive modification to standard multislice—is sufficient to capture high-angle scattering with accuracy comparable to the far more computationally expensive WPM and Klein-Gordon solvers. The WPM and Lanczos methods do not permit coarser z-sampling for a given accuracy level, and the diminishing returns beyond moderate z-refinement make them impractical for routine use.

These results do not diminish the theoretical value of the WPM or full Klein-Gordon solvers: they remain essential as ground-truth references and may prove necessary in extreme regimes not tested here (e.g., very thick high-Z samples at energies below 100 keV). However, for the broad range of conditions encountered in modern TEM—including convergent-beam and frozen-phonon simulations—the angular spectrum multislice method provides an excellent balance of accuracy and computational efficiency.

The differentiable refractive-index binning strategy introduced for the WPM remains a useful contribution, as it enables gradient-based inverse problems such as ptychography to be formulated with a wide-angle forward model when needed. Future work will explore the limits of the angular spectrum approximation in extreme scattering geometries and integrate the method with large-scale ptychographic reconstructions.

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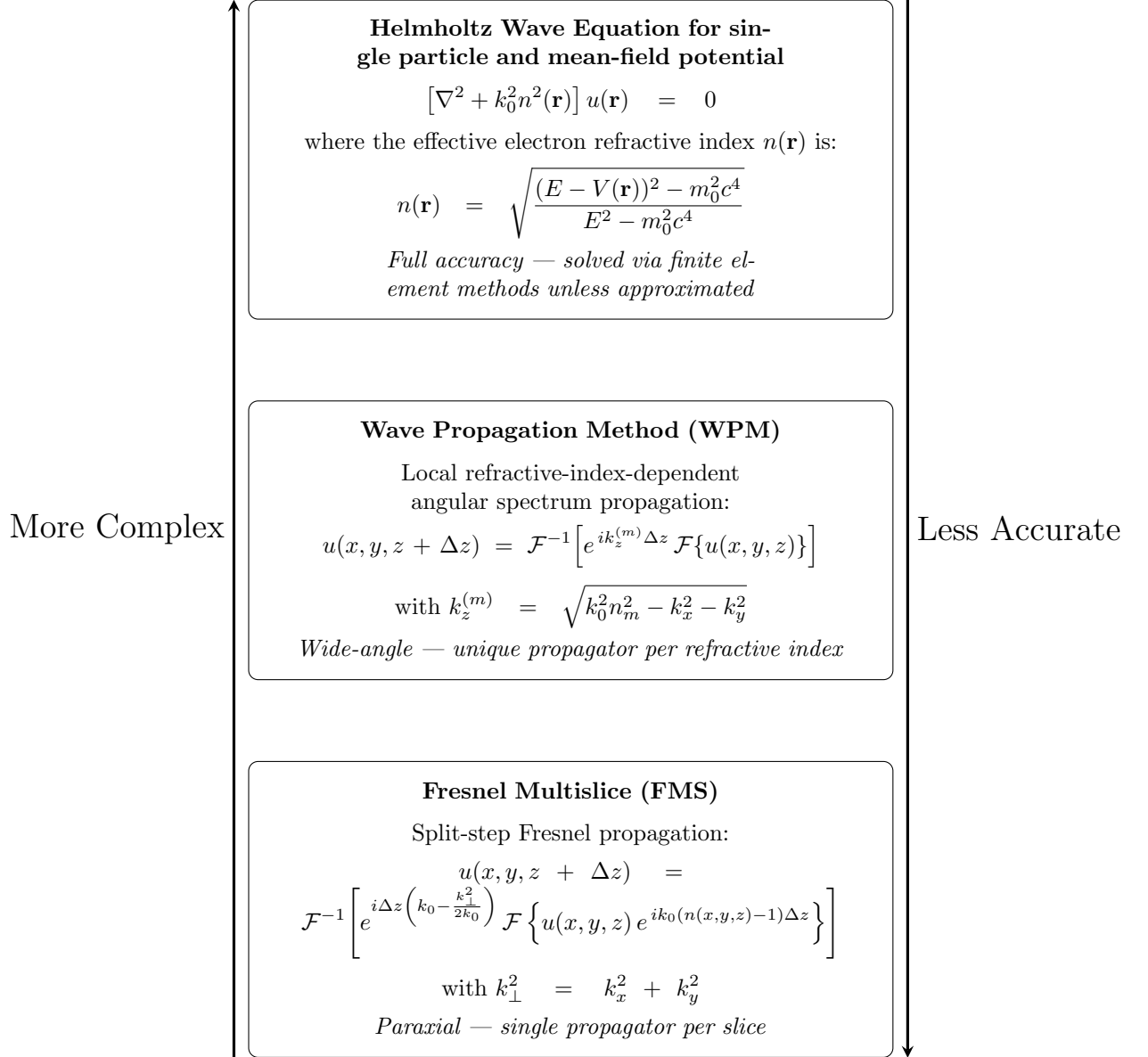


Figure 1: Schematic comparison of the exact Helmholtz formulation, WPM, and Fresnel multislice, illustrating the trade-off between accuracy and computational cost.

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`Au_beam_amplitudes`

Figure 2: Amplitude of the central $[0,0]$ and high-angle $[0,28]$ (135 mrad) diffracted beams as a function of crystal thickness for Au $[100]$ at 300 keV, recreating figure 3 of [3]. The angular spectrum MS (ASMS) closely tracks the WPM and KG ground truth, while Fresnel MS (FMS) diverges at high angles.

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`z_sampling_convergence`

Figure 3: Z-sampling convergence for Au $[110]$ at 200 keV (~ 100 unit cells). Left: relative L^2 error of the diffraction pattern vs. the KG ODE ground truth (128 sl/cell). Right: wall-clock runtime. All methods converge at a similar rate; the more expensive WPM and KG solvers offer no z-sampling advantage.

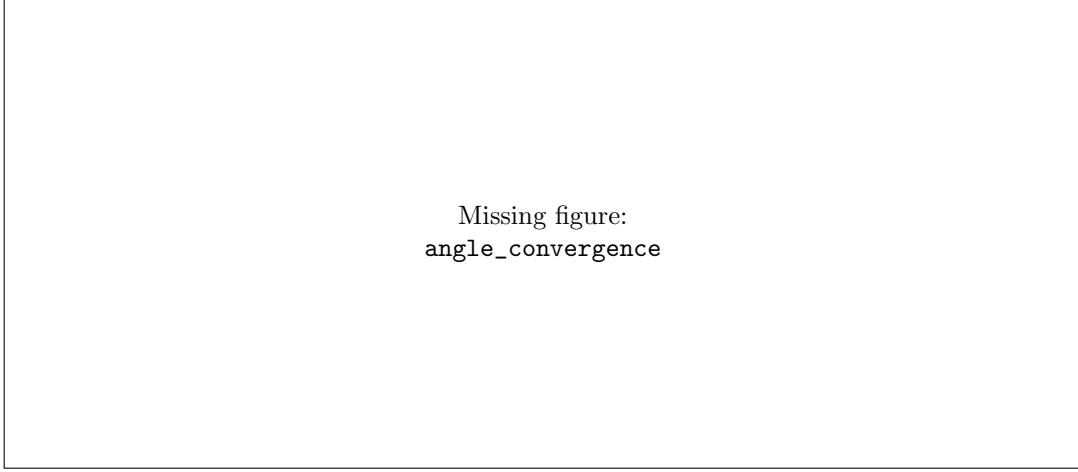


Figure 4: Relative L^2 error of the CBED diffraction pattern versus the KG ODE ground truth (128 sl/cell) as a function of probe convergence semi-angle for Au [110] at 200 keV (32 sl/cell for test methods). The angular spectrum method tracks the WPM and KG FWD accuracy throughout.

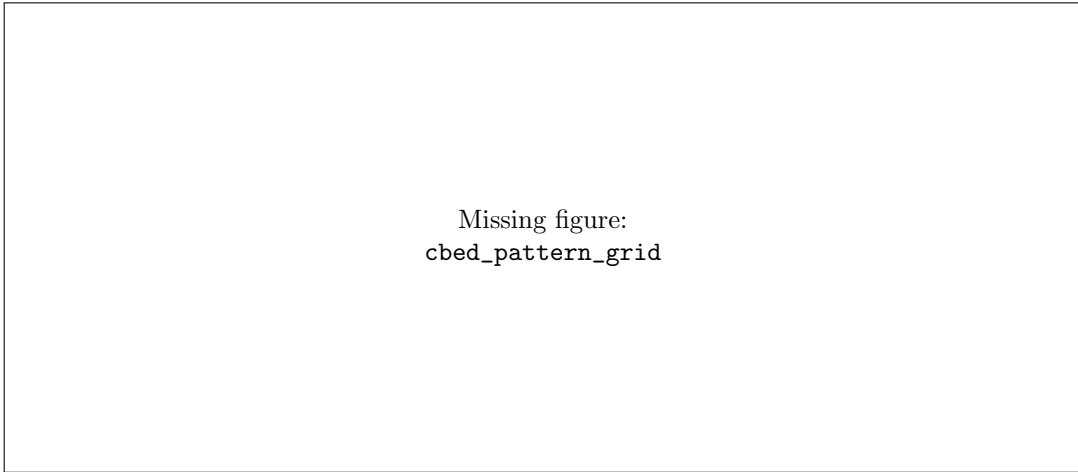


Figure 5: CBED patterns for Au [110] at 200 keV at selected convergence semi-angles (5, 20, 50, 100 mrad). Columns: KG ODE ground truth, Fresnel MS, Angular Spectrum MS, WPM, KG FWD Lanczos. The angular spectrum patterns are visually indistinguishable from the WPM and KG references across all tested semi-angles.

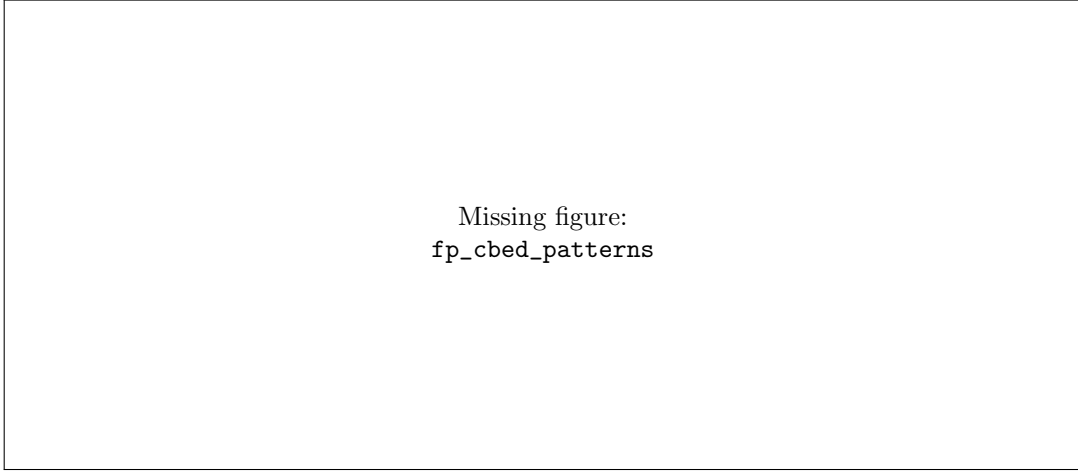


Figure 6: Frozen-phonon averaged CBED patterns for Si [111] at 200 keV (6.2 mrad probe, 16 configurations, 16 sl/cell, 120 unit cells). Intensities displayed on a logarithmic scale.

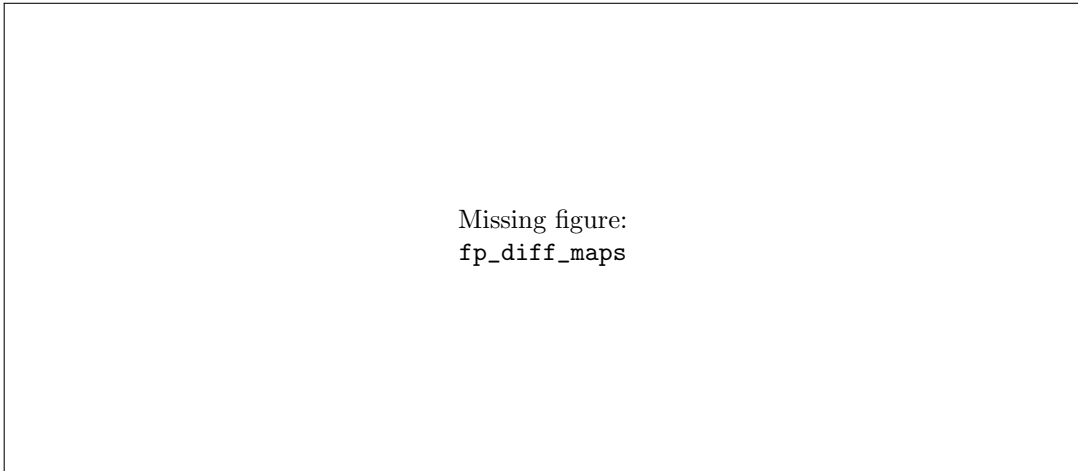


Figure 7: Difference maps relative to KG FWD Lanczos for Si [111] frozen-phonon CBED. Top row: normalised diffraction pattern difference. Bottom row: normalised exit-wave amplitude difference.